CFI Intensified Camera Package Characterization Report

Combustion Integrated Rack (CIR) Cool Flames Investigation (CFI)

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AUTHORIZED by CM when under FORMAL Configuration Control	
Date	**Signature**



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PREFACE

Under the Space Flight Systems Development and Operations Contract (SpaceDOC), the National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation facility for conducting fluid physics and combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Fluids and Combustion Facility (FCF), consists of two test platforms: the Fluids Integrated Rack (FIR) and the Combustion Integrated Rack (CIR). Also included in SpaceDOC are the required support efforts for Mission Integration and Operations, consisting of the Telescience Support Center (TSC) and Mission Integration and Planning (MIP).

The Cool Flames Investigation (CFI) project is a droplet combustion experiment to be executed in the Combustion Integrated Rack (CIR) on the International Space Station (ISS). The overall objective of this proposed research is to conduct further systematic experimental measurements of large droplets that exhibit the cool flame mode of combustion using the Multiuser Droplet Combustion Apparatus (MDCA) currently installed in the CIR. The experiments will examine the hot and cool flame behavior of n-Dodecane, Farnesane, and mixtures of n-Dodecane and Isododecane in varied ambient environments (diluent/oxidizer composition and ambient pressure). Numerical simulations with detailed chemical kinetic models and theoretical analyses with reduced kinetics will complement the experiments with the ultimate goal to improve and validate the chemical kinetic models.

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1.0 INTRODUCTION

The Cool Flames Investigation (CFI) is a new experiment that continues the FLEX and FLEX-2 studies of fuel droplet combustion. This experiment, however, is designed to explore and expand on the recent Cool Flames discovery by the FLEX investigation. Results from FLEX have revealed a two-stage burning event where a heptane droplet that appeared to extinguish actually continues to vaporize without a visible flame. This knowledge from the microgravity study could contribute to reduced pollution and better mileage in engine design, due to improved prediction of flame behavior during combustion.

1.1 Purpose

The purpose of this document is to report the results for characterization of the CFI Flight Intensified Camera Package (67212MFAN16000).

1.2 Scope

The scope of this report covers all characterization results partially derived from implementing the test plan CIR-TST-4388A in February, 2016. These tests characterize the CFI Intensified Camera Package for flight. Based on the results and observations from these tests, any procedural changes, deviations from the plan, and results will be recorded in this final test report. This document does not cover all verification test results.

1.3 Classification

The classification for the CFI Intensified Camera Package is Flight Model.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents referenced herein, the requirements of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. In the event of a conflict between this document and higher-lever documents, the higher-level documents shall take precedence over this document.

All documents used, applicable or reference are to be the issues defined in the current version of the SpaceDOC contract. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The documents in the following table are applicable to the CFI project to the extent specified herein.

67212MFAN16000	Intensified Camera Assembly
CFI-VPLN-002A	Cool Flames Investigation (CFI) Verification Plan
CFI-REQ-001A	Cool Flames Investigation (CFI) Engineering Requirements Document
CIR-TST-4388A	Intensified Camera Package Functional Test Plan
P13001	Procedure for Control of Nonconforming Product

2.3 Reference Documents

The documents in the following table are provided only as reference material for background information and are not imposed as requirements.

SSP-57000	Pressurized Payloads Interface Requirements Document
CFI-CONOPS-003	CFI Concept of Operations

3.0 FACILITIES, EQUIPMENT, AND INSTRUMENTATION

3.1 Test Facilities

The CFI flight Intensified Camera Package Characterization took place from February 5, 2016 through February 12, 2016 in the ZIN Technologies lab facility located at 6749 Engle Road.

3.2 Equipment and Instrumentation

TABLE II lists the name and type of the equipment and instrumentation required for the Intensified Camera Package Characterization. The calibration status of all instrumentation will be verified as appropriate before testing begins. Any changes to or additional support equipment will be added to this table at the time of the characterization procedure.

TABLE I. Equipment and Instrumentation Information

Equipment	Manufacturer	Model Number	Cal Due Date
CIR UML Simulator	ZIN Technologies	67212ETAH20820 S/N: 001-B NASA Tag: 73706	N/A
Dual DC Power Supply, +3.3 VDC / 24 VDC	Tektronix	Model: PS-280 S/N: TW52908 NASA Tag: 12129	N/A (via voltmeter)
DC Regulated Power Supply, 28 VDC	Kenwood	Model: PD36-20AD S/N: 5010016 ZIN Tag: Z000162	N/A (via voltmeter)
Digital Voltmeter/Multi-meter	Fluke	Model: 179 S/N: 81730258 ZIN Tag: Z000371	11 Mar, 2016
GSE Test Cable (67212EGAN16907)	ZIN Technologies	S/N: 1001 NASA Tag: 75983	N/A
IPSUG - Intensified Camera Cable (67212EGAN16903)	ZIN Technologies	S/N: 1001 NASA Tag: 68185	N/A
Serial to RS-422 Converter		MAX3162EVKIT+	N/A
Laptop w/ Serial Port Pro-Motion SW (≥ Ver.3.83)	Dell	Model: Latitude E5510 S/N: CR8W4N1 ZIN Tag: Z004622	N/A
Laptop w/ Ethernet & Serial Port Bobcat GEV Player (≥ Ver.3.1.3) VSI Intensifier SW (Ver.1.0.2.25)	Dell	Model: Inspiron 15-5000 S/N: 3S7NK22 NASA Tag: 68153	N/A
CIR Sapphire Window Assembly (67212MEDH33000)	ZIN Technologies	67212MEDH33000 S/N: 2006	N/A
Integrating Sphere (20”) w/ Controller & Power Supply	Labsphere	Model: SC-5500 S/N: 06561 NASA Tag: 13309	N/A
LED Light Source w/ LEDs (White Light, 455nm)	Lightspeed Technologies	Model: HPLS-36APL750 S/N: 0609520 NASA Tag: 43895	N/A

4.0 DESCRIPTION OF TEST ARTICLE & CONFIGURATION

4.1 Test Article

4.1.1 Intensified Camera Assembly

The Cool Flames Intensified Camera Assembly (67212MFAN16000) consists of three components that make up the entire assembly: the Camera Control Module (67212MFAN16100), the Optics Housing, aka Filter Barrel (67212EFAN21000) and the Camera Sub-assembly.

The Intensified Camera Assembly flight unit was assembled in-house at 6749 Engle Road.

TABLE II. Article

Description	Manufacturer	Drawing Number	S/N
Intensified Camera Assembly	N/A	67212MFAN16000	2001

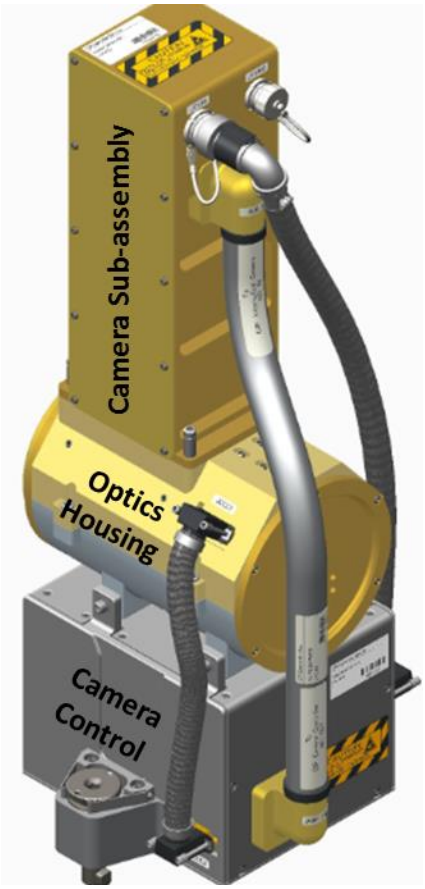


FIGURE 1. Cool Flames Intensified Camera Assembly

4.1.1.1 Camera Control Module (CCM)

The Camera Control Module (CCM) is shown in Figure 2. It serves as the mounting interface to a CIR Universal Mounting Location (UML). The CCM supplies Power Distribution capability, using 28 VDC input from CIR and converts it to 12 VDC required by the camera, intensifier and other components. The CCM also routes forced air cooling provided by the CIR up to the camera, via an attached cooling duct.

Also housed in the CCM are the boards necessary for Stepper Driver and Motion Control functions that are used by the rotating filter barrel. Finally RS-232 and RS-422 communications capabilities are supported by boards housed here as well.



FIGURE 2. Camera Control Module (CCM)

4.1.1.2 Optics Housing / Filter Barrel

The Optics Housing front end is a UV Fused Silica front Window for protection of the internal components. The back end is a 45° turning Mirror with a UV enhanced Al coating, in order to fold the optical path.

Within the Optics Housing (Figure 3) is a stepper motor driven filter barrel with rotational capability. The Filter Barrel has slots for 4 filters.

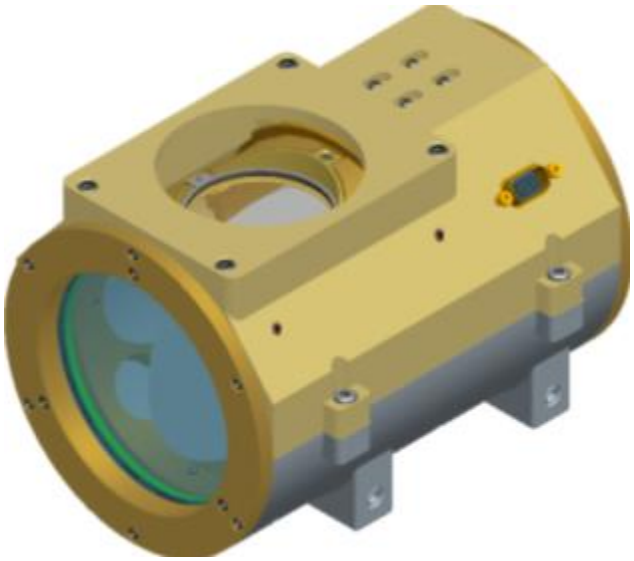


FIGURE 3. Optics Housing / Filter Barrel

Cool Flames has selected the following for the 4 slots: 1 – UV Fused Silica, 2 – CH* emissions (431 nm ± 10 nm), 3 – Formaldehyde emissions (390 nm – 490 nm) and 4 – Empty. Transmission curves for the various filters are shown in Figures 4 – 6.

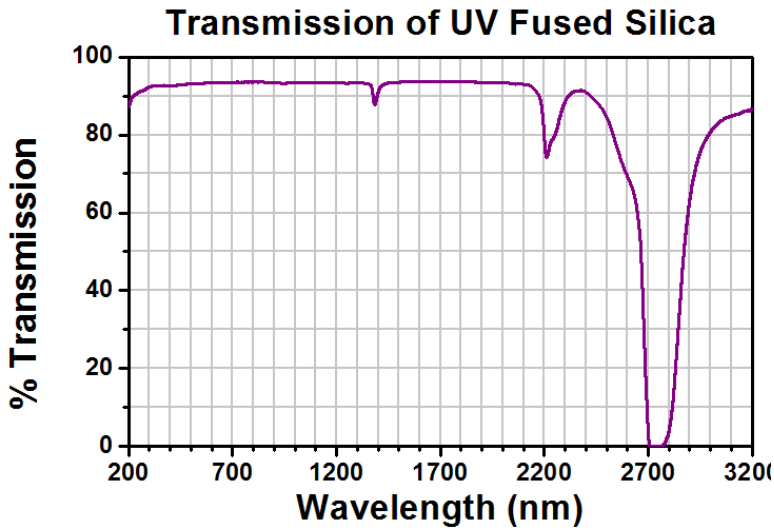


FIGURE 4. UV Fused Silica Transmission

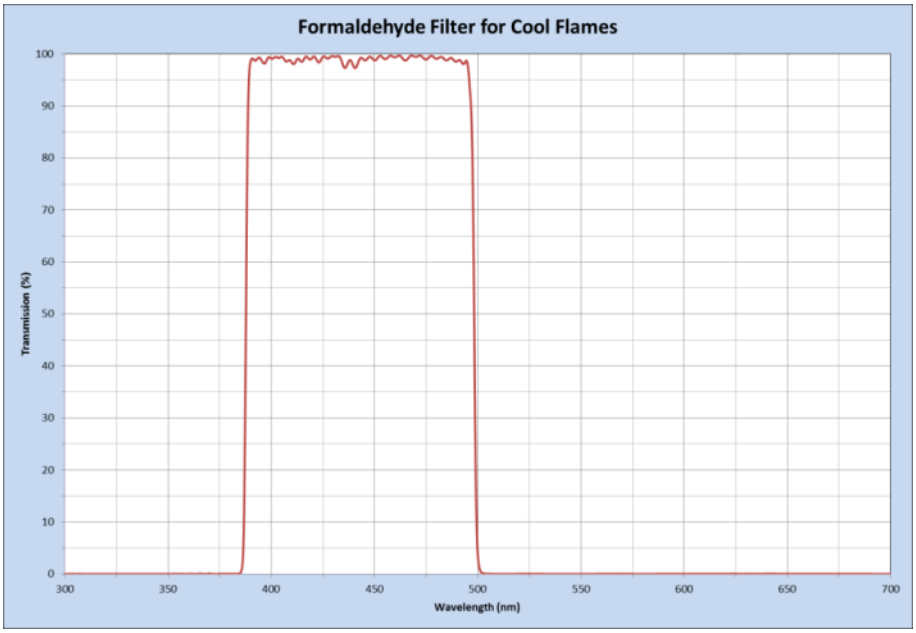


FIGURE 5. Formaldehyde Filter Transmission

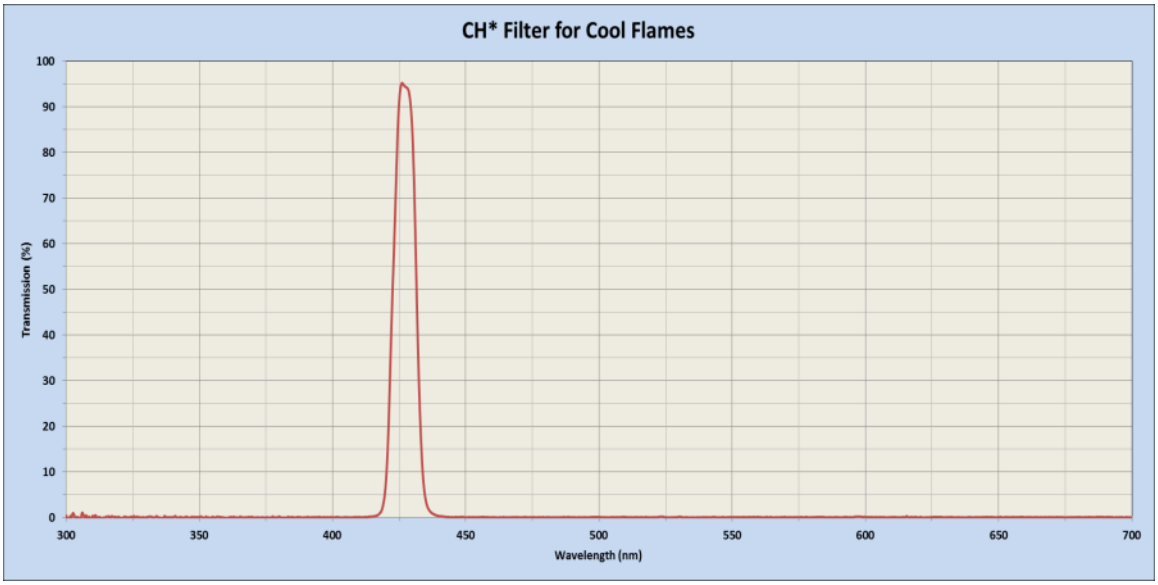


FIGURE 6. CH Filter Transmission

4.1.1.3 Intensified Camera Sub-assembly

The Camera Sub-assembly includes the base camera (Imperx) with Gen-III Enhanced Blue Intensifier integrated by Video-Scope, and a UV Quartz 50 mm Lens.

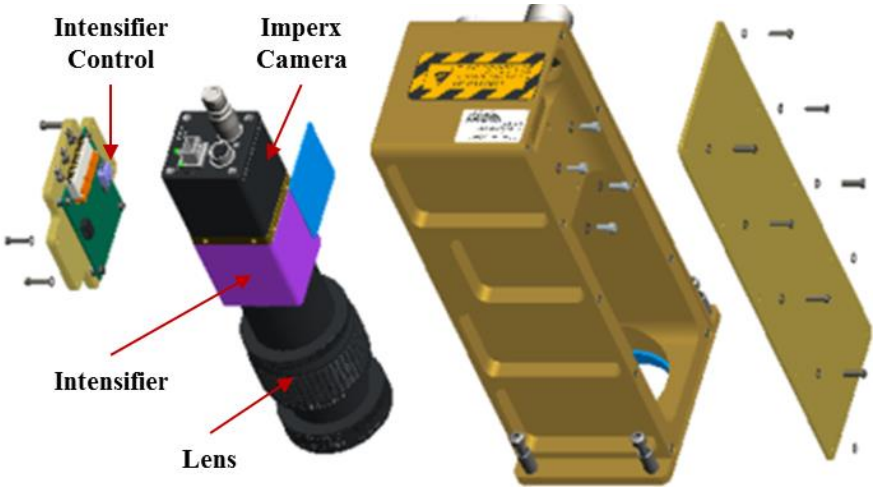


FIGURE 7. Camera Sub-assembly

The Camera Sub-assembly includes the base camera (Imperx) with Gen-III Enhanced Blue Intensifier integrated by Video-Scope, and a UV Quartz 50 mm Lens.

4.1.1.4 Test Configuration

The Radiometer Assembly to be calibrated and the Blackbody Source are positioned on an optical bench. The Blackbody Source output is aligned, via optical mounting hardware, with the opening of the particular Radiometer Package (BB or NB) being tested.

The optical mounting hardware allows for alignment in both the X and Y directions and sets the distance from the detector input of each Radiometer to the Blackbody aperture wheel (Z direction), equal to the appropriate Flight distance of 110 mm. See FIGURE 5.

FIGURE 8. CFI Radiometer & Blackbody Source Alignment

During previous testing performed in 2011, it was observed that the fixed aperture wheel that was directly mounted on the Blackbody source heated up over time and actually emitted radiation at significant levels when compared to radiation from the Blackbody cavity. This extra and unintended radiation was shown to contribute to the total Radiometer signal, resulting in systematic error due to increasing the effective aperture size.

To reduce this error, all Blackbody characterizations in this document were run with the setup shown in FIGURE 6. A simple modification removed the fixed Blackbody aperture wheel and mounted it on a slide rail, allowing the shield to be retracted away from the hot Blackbody source between measurements. By limiting the time resident near the hot Blackbody source, the aperture wheel could be held at room temperature to within a few tenths of a °C. Note that in this configuration the aperture wheel is still located the same distance from the Blackbody source as it was in its original fixed position, thus continuing to allow the aperture wheel holes to be the defining emitters in the system, to first order.

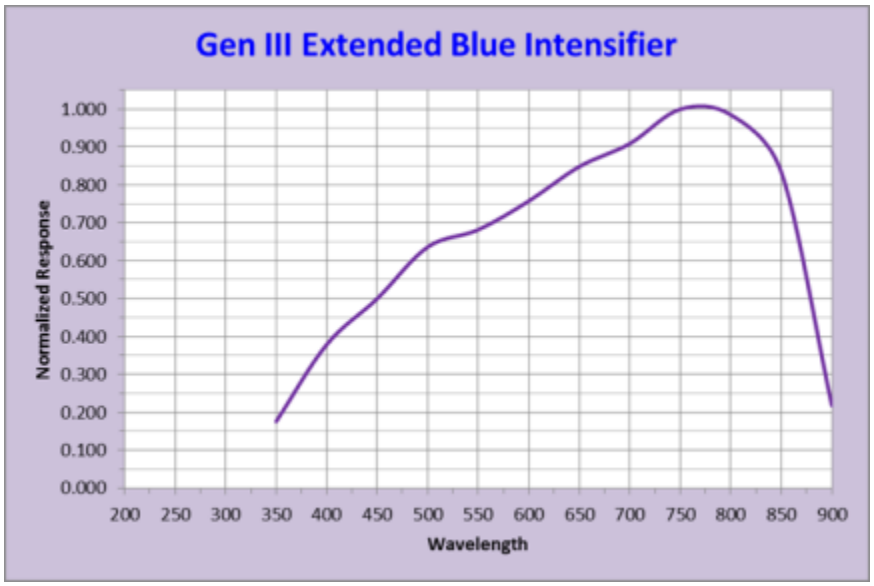


FIGURE 9. Gen-III Extended Blue Response Curve

The Radiometer is connected to the ±12 VDC Power Supply (for power in) via the Radiometer Test Cable (67235EGAC13966) and to a voltmeter and oscilloscope to monitor the output signal V_{out}. Refer to FIGURE 7 for the connection schematic. An additional voltmeter was connected to the Power Supply to ensure the input voltages remained within the ±12 VDC required.

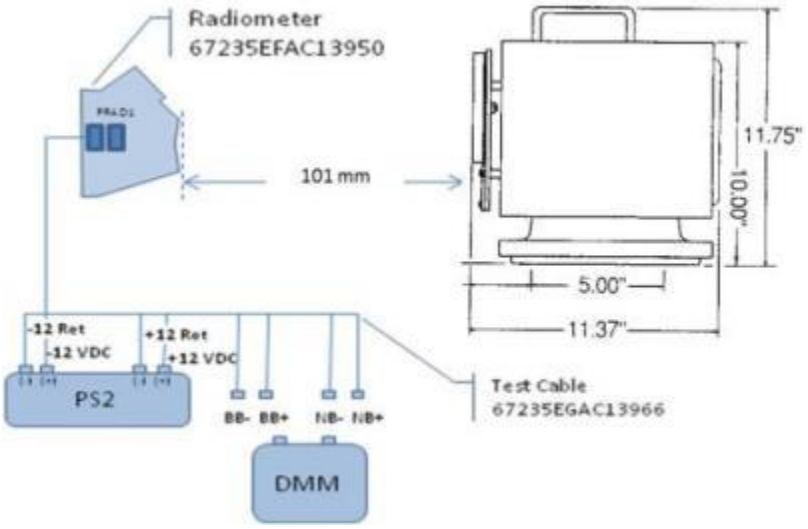


FIGURE 10. Cabling Schematic for Characterization Setup

The DC Power Supply, Voltmeters, Oscilloscope and Blackbody Temperature Controller are located on a bench adjacent to the Radiometer and Blackbody (on the optical bench behind this shelf). FIGURE 8 shows a photo of the support equipment in the laboratory setting.

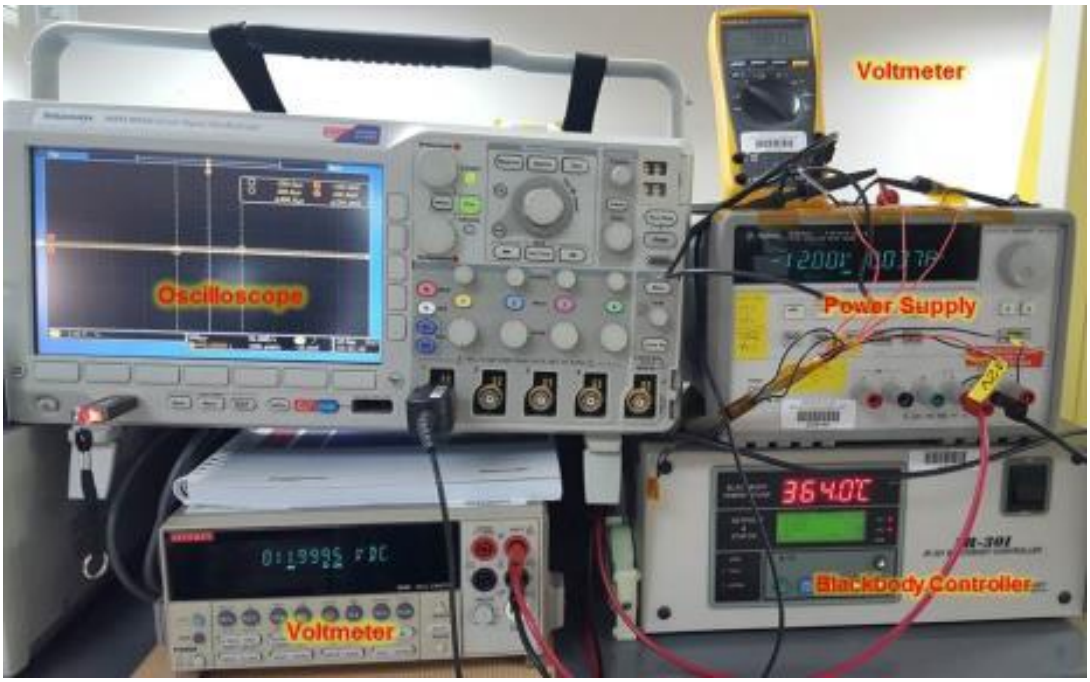


FIGURE 11. CFI Radiometer Support Equipment

5.0 TEST PROCEDURE

These procedures cover three characterizations of the CFI Radiometers:

i) Radiometric characterization via Blackbody Source

ii) Field of View (FOV) roll-off measurement or instrument function

iii) Signal to Noise Ratios (SNR) of the radiometer subsystems

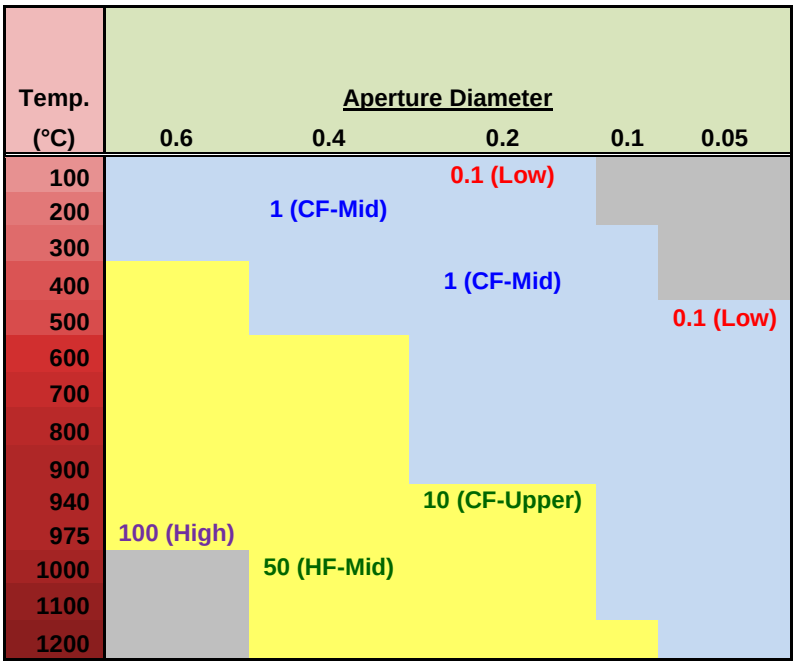
5.1.1 Radiometer Characterization (performed separately on BB & NB)

The Radiometer Characterization Procedure is performed separately for each of the Broadband and Narrowband Radiometers using the "Radiometer Characterization Procedure Datasheet" in APPENDIX B.

Within each of the Radiometer Characterization Procedures, specific data must be recorded for the Radiometer voltage output over a range of apertures using a separate APPENDIX C, "Radiometer Characterization Data Collection Sheet" for each blackbody temperature. This simulates various source Irradiance levels covering the operating range. To best align with the test procedures, a unique "Radiometer Characterization Data Collection Sheet" is used for each temperature and radiometer.

Table III shows the test matrix for the characterization of the two Radiometers. Each radiometer is tested separately, and aligned normal to the blackbody source. The output voltage is recorded at each temperature and aperture value in the table. To gain some redundancy in the characterization data, the test matrix spans the envelope of the Low and High required irradiance values, as documented in CFI-REQ-0001, as well as the Midpoint and Upper end of the Cool Flames region and the overall Midpoint. These points are indicated in TABLE III.

TABLE III. Radiometric Characterization Test Matrix



5.1.2 FOV Roll-off Characterization

The instrument function of the radiometer package is measured over a range of off-axis alignment for each radiometer. Each Radiometer is centered and aligned normal to the blackbody source, at a distance of 11.0 cm from the blackbody aperture to the Radiometer TO-5 can face. The Radiometer is translated in the horizontal (X) direction over a range of ±70 mm about this center, at 10 mm increments, while recording the output signal level. This determines the X roll-off characteristics.

The vertical (Y) direction characterization is measured by raising/lowering the Radiometer on a lab jack over ±70 mm to measure the Y roll-off characteristics, while the Radiometer remains at its X axis center point. The Field of View Characterization Procedure (APPENDIX D) and FOV Characterization Data Collection Sheets (APPENDIX E) are used for this characterization.

5.1.3 Signal to Noise Ratio Characterization

The signal-to-noise characteristics of the radiometers are measured at Low (close to minimum) signal level, Medium signal level, and High signal level. The time sequence of samples is measured by an oscilloscope sampling AC and DC at 25 kHz. The DC input is the signal, and the AC input is the noise in the sampling bandwidth. The AC sampling is smoothed over 100 Hz to represent the flight acquisition sample time. Data are logged on the SNR Data Collection Sheet (APPENDIX F).

5.1.4 Response Time Characterization

The response time of the BB and NB detectors are measured using the calibrated blackbody source with a chopper wheel to provide a step function input at a frequency of 2Hz. The output voltage is analyzed to calculate the rise time from 10% to 90% of the maximum signal response.

5.2 Data Collection and Reporting

All data are collected and recorded on the attached data sheets. Data sheets will be maintained to document the history of the CFI Radiometer Characterization. The data sheets include, but are not limited to, the following information:

1. Date and time
2. Identity of individual making the entry
3. Environmental conditions
4. Performance parameter measurements
5. Power ”On” Time
6. Failure observations
7. Configuration identification
8. Record of non-conformances, repair / maintenance actions, etc.

6.0 DATA ANALYSIS

6.1 Imaging Field of View (FOV)

This section outlines the test setup, procedure and analysis for the test(s) to determine the Field of View (FOV) of the CFI Intensified Camera Package. The Field of View results are summarized at the end.

Per CIR-VPLN-4369, Requirement 5.1.3: The Field of View (FOV) for the Intensified Camera shall be a minimum of 8.0 cm, at a distance of 253.2 mm ± 2 mm from the object plane.

An 8.0 cm FOV is most commonly defined as a circle with an 8.0 cm diameter. FOV can also be referenced in terms of being projected from a square or rectangular CCD, in which case both horizontal and vertical dimensions would be at least 8.0 cm.

6.1.1 FOV Test Setup

The CFI Intensified Camera Assembly was mounted on the CIR UML Simulator tub and connected to the specified power supplies and laptops for power, cooling and control capabilities.



FIGURE 12. Concentric Squares Target

The Concentric Squares target (Figure 13) was placed at the specified Object Distance = 253.2 mm from the outer surface of the Assembly using the Z translation. The Target was centered in the image using the X & Y translation adjustments. The 20” Integrating Sphere exit port was placed just beyond the target, with the target centered within the port. The source was a 455 nm LED, set to output ~23 foot-Lamberts. See the complete setup in Figure 14.

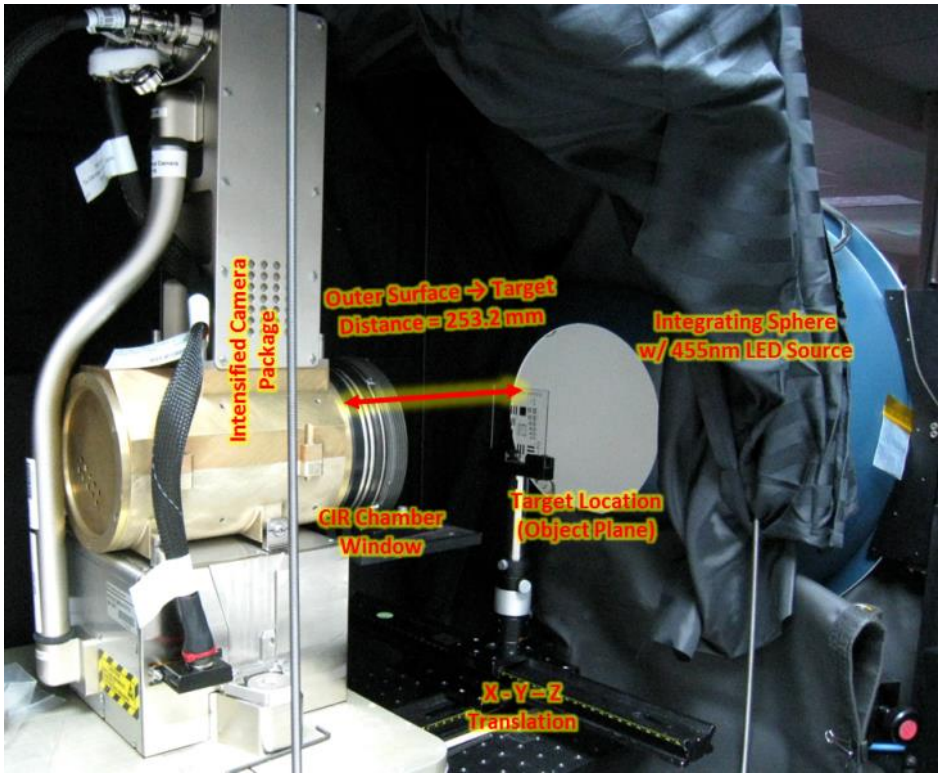


FIGURE 13. Intensified Camera Characterization Setup

Intensified Camera Settings used for this test are summarized in the following table:

TABLE IV. Intensified Camera Settings – FOV

Imperx Camera Functions			Intensifier Functions		
Binning (# pixels)	Width	1x (1392)	Gain Amp.	(2500 - 80000)	2500
	Height	1x (1040)	Gate Mode	(DC/Pulsed)	DC
Frame Rate (Camera Speed)	Normal or Overclocked	30 fps	Pulse Width	Safe = 0.020 µs	n/a
Pixel Format (Bit Depth)	16 Bit	Mono16			

The Field of View measurement is independent of Filter selection, thus the Formaldehyde filter was used as the default for this test.

6.1.2 FOV Test Data and Analysis

An image of the Concentric Squares target was captured ("Squares_Form_455") using the Setup and Camera/Intensifier parameters specified in the previous section.

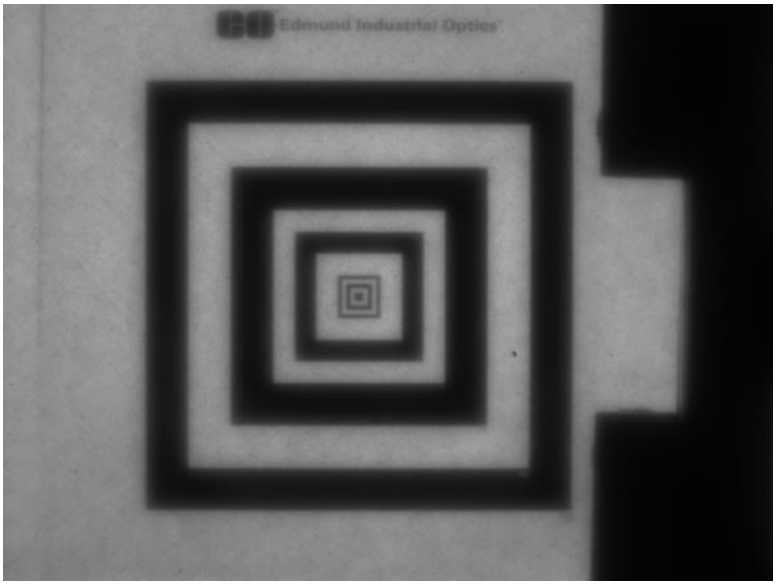


FIGURE 14. Intensified Camera FOV Image "Squares_Form_455"

The image was opened using Spotlight-16 (Ver. 2004.8.27) imaging software. Using the "Scale" tool, the line length (in pixels) was stretched to span the outermost black square on the target, which equals 50mm. Using that scaled value, the "Line Tool" was then stretched to span the entire image in both the vertical and horizontal directions in order to determine the overall Field of View (FOV). This was repeated to establish an average value in each direction, see the Table below:

TABLE V. Intensified Camera Results – FOV

FOV			
Pixels (Width)	1392		
Pixels (Height)	1040	Pixels (Diag)	1737.6
Width	90.21		
Height	67.38	Diag	112.596
Width	89.84		
Height	67.1	Diag	112.132
Width	90.68		
Height	67.73	Diag	113.182
Width	90.21		
Height	67.38	Diag	112.596
Avg. Width	**90.24**		
Avg. Height	**67.40**	Diag	**112.63**

6.1.3 FOV Test Results

The **Field of View** determined by the averaged measurements is: **FOV = 90.24 mm x 67.40 mm.**

This meets the 80 mm requirement in the horizontal axis, but falls slightly short by 12.6 mm in the vertical axis, as shown in the graphic in Figure 16.

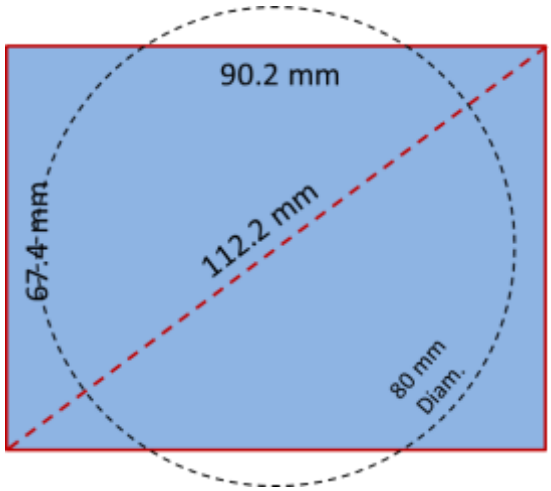


FIGURE 15. CFI Intensified Camera Field of View (FOV)

Thus, based on the Field of View requirements, the Intensified Camera does not comply. However with approved waiver: CIR-WAI-xxxx, the requirement is met.

6.2 Imaging Spatial Resolution

This section outlines the test setup, procedure and analysis for the test(s) to determine the Spatial Resolution of the CFI Intensified Camera Package. The Spatial Resolution results are summarized at the end.

Per CIR-VPLN-4369, Requirement 5.1.5: The Spatial Resolution for the Intensified Camera, at the object plane, shall be a minimum of 100 μm (≥ 5 lp/mm) over the entire FOV.

Spatial Resolution is the imaging system's ability to determine object detail. When referring to CCD image sensors can be simplistically defined in terms of object size divided by the number of pixels in the sensor. But that doesn't consider the overall performance of the optical system. The most effective means of specifying total performance is the Modulation Transfer Function (MTF), which incorporates image contrast along with resolution. In terms of an optical system, MTF is typically defined as a percentage of the object contrast that is reproduced in the image. A typical Contrast performance limit is 10% and since MTF was not specified, 10% cutoff was used for this requirement.

6.2.1 Spatial Resolution Test Setup

The CFI Intensified Camera Assembly was mounted on the CIR UML Simulator tub and connected to the specified power supplies and laptops for power, cooling and control capabilities.

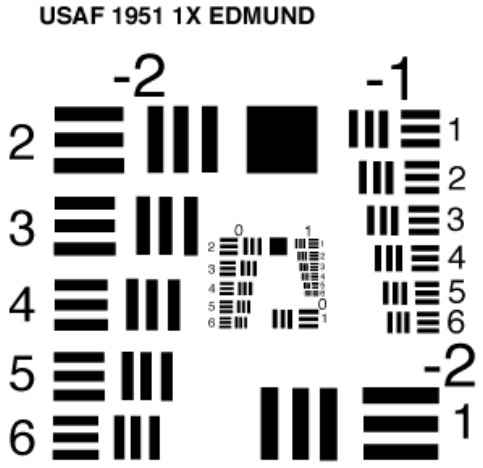


FIGURE 16. USAF Target

The USAF target (Figure 17) was placed at the specified Object Distance = 253.2 mm from the outer surface of the Assembly using the Z translation. The Target was centered in the image using the X & Y translation adjustments. The 20" Integrating Sphere exit port was placed just beyond the target, with the target centered within the port. The source used was a 455 nm LED, set to output ~23 foot-Lamberts. See the complete setup in Figure 18.

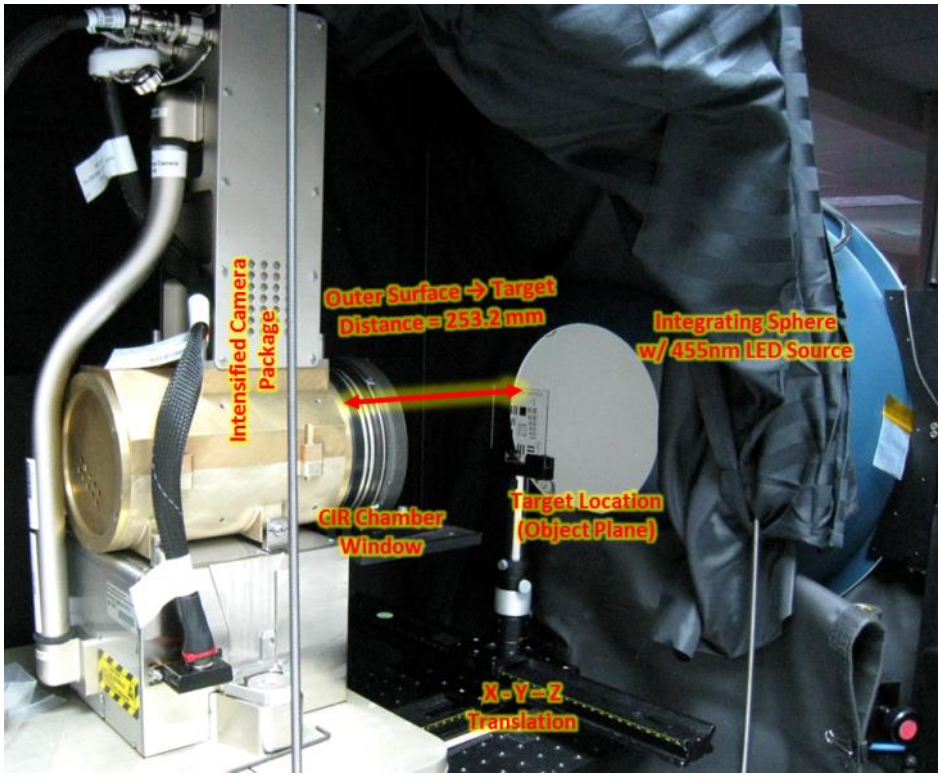


FIGURE 17. Intensified Camera Characterization Setup

Intensified Camera Settings used for this test are summarized in the following table:

TABLE VI. Intensified Camera Settings – Spatial Resolution

Imperx Camera Functions			Intensifier Functions		
Binning (# pixels)	Width	1x (1392)	Gain Amp.	(2500 - 80000)	2500
	Height	1x (1040)	Gate Mode	(DC/Pulsed)	DC
Frame Rate (Camera Speed)	Normal or Overclocked	30 fps	Pulse Width	Safe = 0.020 µs	n/a
Pixel Format (Bit Depth)	16 Bit	Mono16			

The Spatial Resolution measurement is dependent on Filter selection, thus all three of the filters (Formaldehyde, CH, & Fused Silica) were used for this test.

6.2.2 Spatial Resolution Test Data and Analysis

Images of the USAF target were captured for each filter ("USAF_Form_455", "USAF_CH_455" & "USAF_FS_455") using the Setup and Camera/Intensifier parameters specified in the previous section.

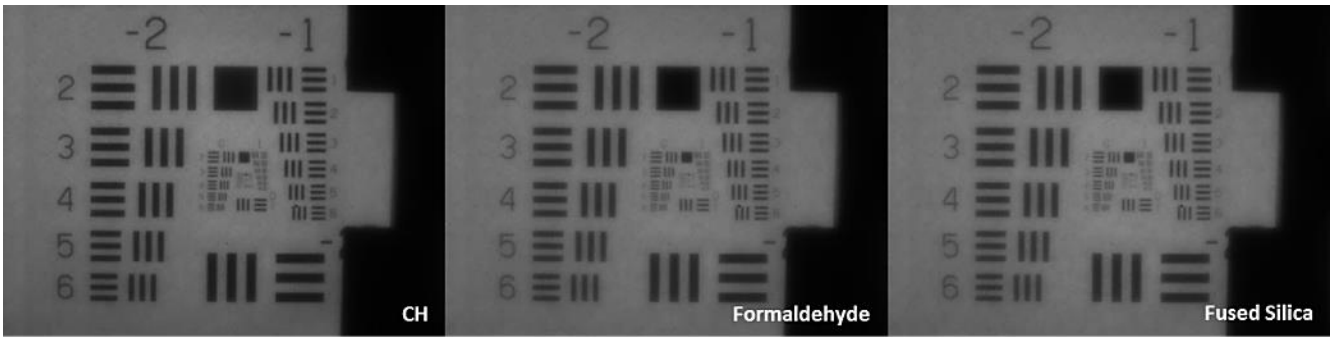


FIGURE 18. Intensified Camera USAF Images (3 Filters)

The images were opened using Spotlight-16 (Ver. 2004.8.27) imaging software.

Using the USAF Target, the resolution limit is the last group of 3 bars that can be seen and distinguished as such, in both horizontal and vertical directions. Breaks within a bar or the overall pattern are acceptable as long as the 3 bars can still be identified.

The CFI resolution requirement is 5 lp/mm (100 μm), which corresponds to USAF Target Group 2, Element 3. From simply viewing the images, it was obvious that the 3 Bars in USAF 2, 3 could not be distinguished from one another in either the horizontal or vertical direction.

From visual inspection of the images, it was determined that the limit where 3 Bars were distinguishable was USAF Group 1, Element 6, corresponding to 3.6 lp/mm (140 μm). Thus the subsequent CFI analysis was performed on Group 1, Element 6.

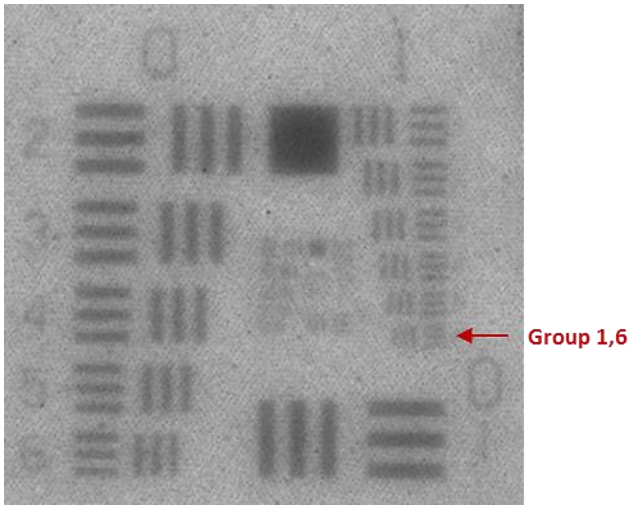


FIGURE 19. Zoom in – USAF Groups 0 & 1

Using the "Rectangular AOI" tool, the Background Mean gray value for each image was measured by centering the AOI within the large black square between USAF Group -2 and Group -1. The "Statistics" function was then used to display Min, Max, Mean & Std. Dev. gray levels (GL) within the AOI. The Mean GL was recorded for each image.

For the contrast measurement, select the desired Group and Element (1,6) and use the "Line Tool" to span the 3 Bars and 2 Spaces in both the horizontal and vertical directions for each image. The Min, Max, Mean & Std. Dev. values are displayed. Contrast is calculated using the following equation:

$$\%Contrast = \frac{Max - Min}{Max + Min - Bkgd_{Mean}}$$

The "Max" values correspond to the black bars and the "Min" values correspond to the spaces in between. This was repeated on multiple images for each of the three filters, to establish an average Contrast for each direction. See the Table below for results:

TABLE VII. Intensified Camera Results – Spatial Resolution (Group 1,6)

Formaldehyde	Bkgd Mean	Min	Max	% Contrast
Horizontal Group 1,6	2827	23976 24616 24600	31392 31043 31207	
		24397.3	**31214.0**	**12.91**
Vertical Group 1,6	2827	24548 25080 25356	31292 31135 30499	
		24994.7	**30975.3**	**11.25**
CH	**Bkgd Mean**	**Min**	**Max**	**% Contrast**
Horizontal Group 1,6	5140	22087 22367 22487	28140 28892 28220	
		22313.7	**28417.3**	**13.39**
Vertical Group 1,6	5140	22347 23104 22815	29183 27524 28568	
		22755.3	**28425.0**	**12.31**
Fused Silica	**Bkgd Mean**	**Min**	**Max**	**% Contrast**
Horizontal Group 1,6	2570	25800 26084 26232	32876 32520 32896	
		26038.7	**32764.0**	**11.96**
Vertical Group 1,6	2570	24996 25708 25776	31372 31484 31648	
		25493.3	**31501.3**	**11.04**

In addition, for the Formaldehyde filter, the full Modulation Transfer Function (MTF) curve was calculated by measuring the Contrast across the range of USAF Target groups. The resulting MTF curve is shown in Figure 21, along with the location of the Intensified Camera resolution limit.

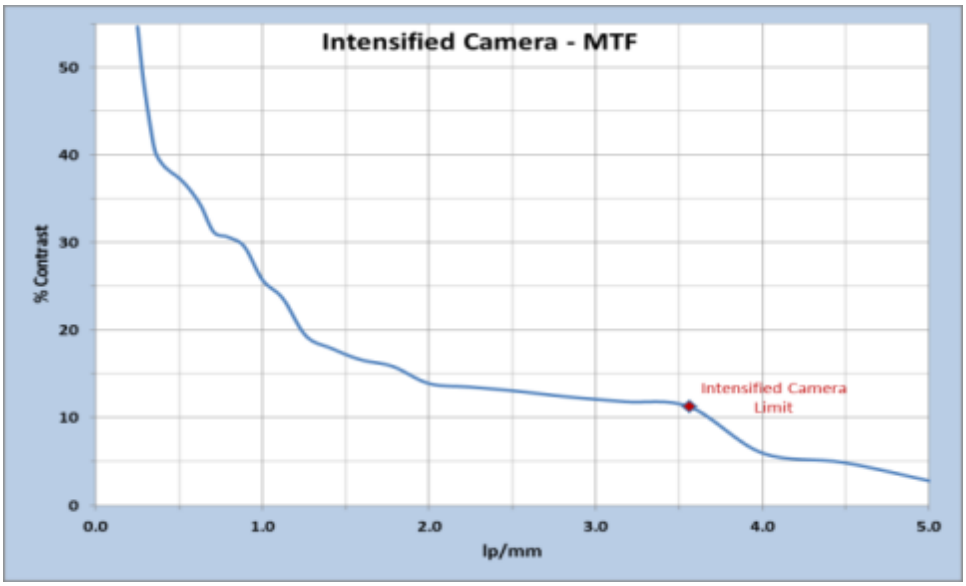


FIGURE 20. MTF – Intensified Camera w/ Formaldehyde Filter

6.2.3 Spatial Resolution Test Results

The Spatial Resolution for each of the three filters in the Intensified Camera Package was determined by the averaged measurements, in both horizontal and vertical directions, of USAF Target Group 1, Element 6, which represented the Resolution limit at just above 10% Contrast:

Formaldehyde (Cool Flame): **3.6 lp/mm (140 µm) @ %Contrast 12.9% (H) x 11.3% (V)**

CH (Hot Flame): **3.6 lp/mm (140 µm) @ %Contrast 13.4% (H) x 12.3% (V)**

Fused Silica: **3.6 lp/mm (140 µm) @ %Contrast 12.0% (H) x 11.0% (V)**

Based on the 5 lp/mm (100 µm) requirement (corresponding to USAF Target Group 2, Element 3), the Spatial Resolution falls slightly short for each of the filters by ~ 1.4 lp/mm (40 µm).

Thus, per the stated Spatial Resolution requirements, the Intensified Camera does not comply. However with approved waiver: CIR-WAI-xxxx, the requirement is met.

6.3 Imaging Depth of Field

This section outlines the test setup, procedure and analysis for the test(s) to determine the Depth of Field (DOF) of the CFI Intensified Camera Package. The DOF results are summarized at the end.

Per CIR-VPLN-4369, Requirement 5.1.4: The Depth of Field for the Intensified Camera shall be a minimum of 3.0 cm.

Depth of Field refers to the range of distance, centered at the object plane that an image appears in focus. It is often defined by Spatial Resolution values at the various stages of depth. If accompanying Resolution values are not given at various points throughout the range, as in this case, it is assumed that the Spatial Resolution measured at the object plane will hold throughout the Depth of Field.

6.3.1 Depth of Field Test Setup

The CFI Intensified Camera Assembly was mounted on the CIR UML Simulator tub and connected to the specified power supplies and laptops for power, cooling and control capabilities.

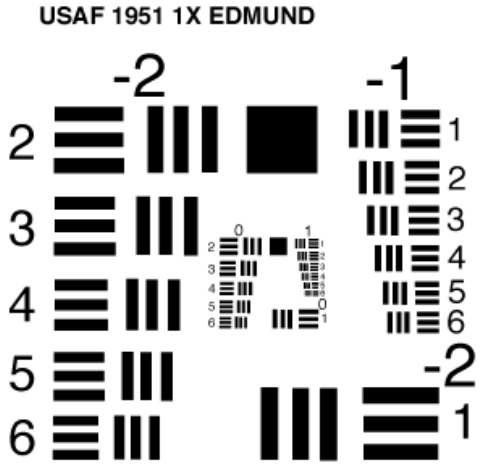


FIGURE 21. USAF Target

The USAF target (Figure 21) was placed at the specified Object Distance = 253.2 mm from the outer surface of the Assembly using the Z translation. This is the "Center" position for the Depth of Field measurements. The 20" Integrating Sphere exit port was placed beyond the target, with the target centered within the port. The source used was a 455 nm LED, set to output ~23 foot-Lamberts. See the complete setup in Figure 22.

For Depth of Field measurements, the optical rail was used for depth translation both toward the camera ("–" direction) and away from the camera ("+" direction).

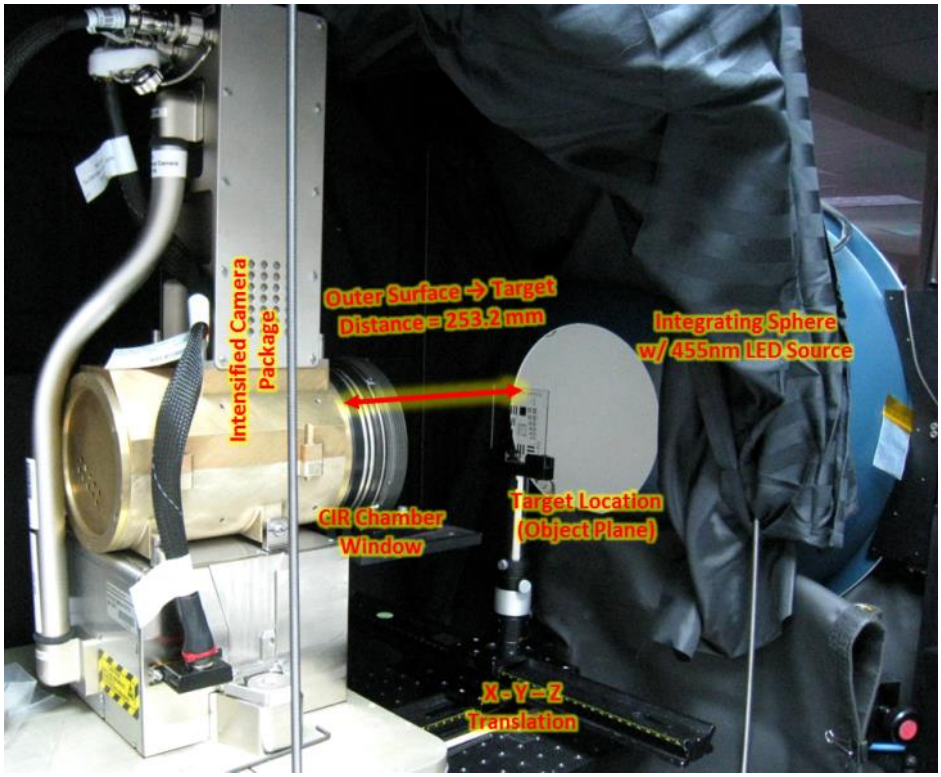


FIGURE 22. Intensified Camera Characterization Setup

Intensified Camera Settings used for this test are summarized in the following table:

TABLE VIII. Intensified Camera Settings – Depth of Field

Imperx Camera Functions			Intensifier Functions		
Binning (# pixels)	Width	1x (1392)	Gain Amp.	(2500 - 80000)	2500
	Height	1x (1040)	Gate Mode	(DC/Pulsed)	DC
Frame Rate (Camera Speed)	Normal or Overclocked	30 fps	Pulse Width	Safe = 0.020 µs	n/a
Pixel Format (Bit Depth)	16 Bit	Mono16			

The Depth of Field measurement is based on Spatial Resolution measurements, which are dependent on Filter selection, thus all three of the filters (Formaldehyde, CH, & Fused Silica) were used for this test.

6.3.2 Depth of Field Test Data and Analysis

For each of the 3 filters, images of the USAF target were captured at the Center position (253.2 mm distance). The USAF target was then translated along the optical rail away from the camera to the +5 mm, +10 mm and +15 mm positions, with images taken at each of the positions. Then the USAF target was translated past the center position, toward the camera to the -5 mm, -10 mm and -15 mm positions, again taking images at each position. Examples of images at all 7 positions are shown below for the Formaldehyde filter. Similar sets of images were collected for the CH and Fused Silica filters as well.

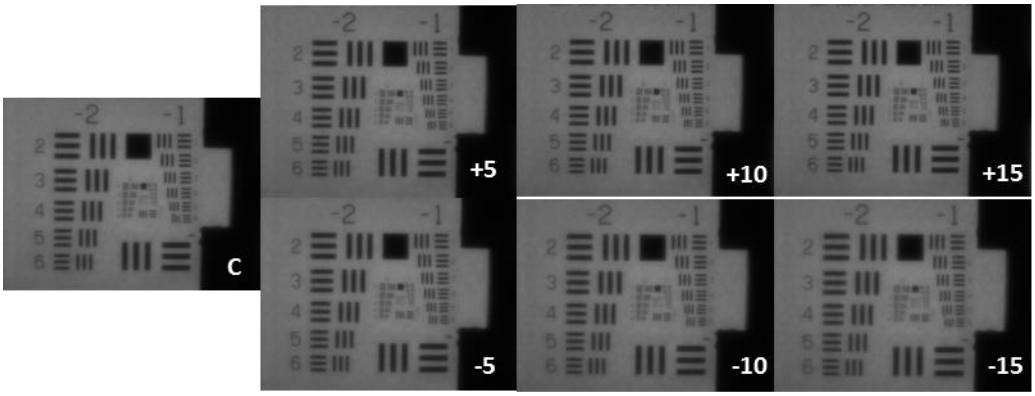


FIGURE 23. Intensified Camera DOF Images (Formaldehyde Filter)

The images were opened using Spotlight-16 (Ver. 2004.8.27) imaging software.

Using the USAF Target, the resolution limit is the last group of 3 bars that can be seen and distinguished as such, in both horizontal and vertical directions. Breaks within a bar or the overall pattern are acceptable as long as the 3 bars can still be identified.

Based on the Spatial Resolution results, and from visual inspection of the images, it was determined that the limit where 3 Bars were distinguishable was USAF Group 1, Element 6, corresponding to 3.6 lp/mm (140 μm). Thus the subsequent Depth of Field analysis was performed on Group 1, Element 6, which represents the baseline Spatial Resolution for this system.

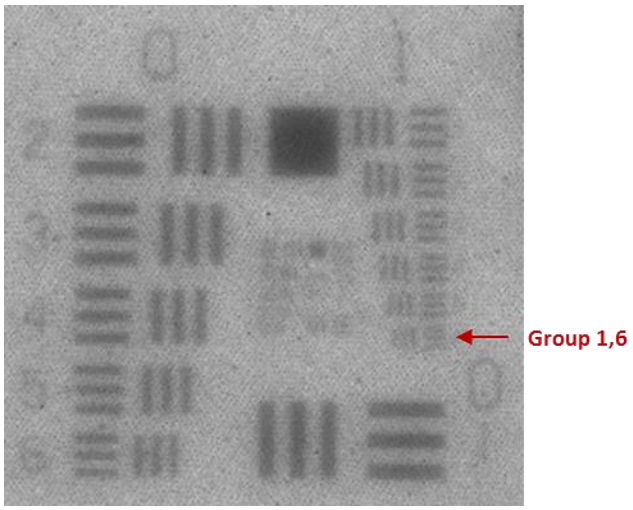


FIGURE 24. Zoom in – USAF Groups 0 & 1 at Center Position

Using the "Rectangular AOI" tool, the Background Mean gray value for each image was measured by centering the AOI within the large black square between USAF Group -2 and Group -1. The "Statistics" function was then used to display Min, Max, Mean & Std. Dev. gray levels (GL) within the AOI. The Mean GL was recorded for each image.

For the contrast measurement, select the desired Group and Element (1,6) and use the "Line Tool" to span the 3 Bars and 2 Spaces in both the horizontal and vertical directions for each image. The Min, Max, Mean & Std. Dev. values are displayed. Contrast is calculated using the following equation:

$$\%Contrast = \frac{Max - Min}{Max + Min - Bkgd_{Mean}}$$

The "Max" values correspond to the black bars and the "Min" values correspond to the spaces in between. This was repeated on multiple images for each of the three filters, to establish an average Contrast for each direction. See the following Table for results:

TABLE IX. Intensified Camera Results – Depth of Field (Group 1,6)

Formaldehyde

Center	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	24397	31214	12.91
Vertical	2827	24995	30975	11.25
+5	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	24368	31352	13.20
Vertical	2827	24112	30379	12.13
-5	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	23748	30487	13.11
Vertical	2827	24184	29932	11.21
+10	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	24112	30295	11.99
Vertical	2827	24384	30331	11.46
-10	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	24724	30843	11.60
Vertical	2827	25052	30675	10.63
+15	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	23456	28232	9.77
Vertical	2827	24672	28280	7.20
-15	Bkgd Mean	Min	Max	% Modulation
Horizontal	2827	25204	28028	5.60
Vertical	2827	26156	28484	4.49

CH

Center	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	22314	28417	13.39
Vertical	5140	22755	28425	12.31
+5	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	23768	30295	13.34
Vertical	5140	24272	29796	11.29
-5	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	24332	30335	12.12
Vertical	5140	24616	30427	11.64
+10	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	24486	30507	12.08
Vertical	5140	24260	30132	11.92
-10	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	23948	31252	14.59
Vertical	5140	24908	30959	11.93
+15	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	24100	28652	9.56
Vertical	5140	23680	27772	8.84
-15	Bkgd Mean	Min	Max	% Modulation
Horizontal	5140	23664	28816	10.88
Vertical	5140	24768	29300	9.26

Fused Silica

Center	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	26039	32764	11.96
Vertical	2570	25493	31501	11.04
+5	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	25036	31636	12.20
Vertical	2570	25740	32092	11.49
-5	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	25664	31568	10.80
Vertical	2570	25652	31408	10.56
+10	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	25436	31768	11.59
Vertical	2570	25816	31884	11.01
-10	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	25912	32280	11.45
Vertical	2570	25840	31992	11.13
+15	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	24816	29644	9.30
Vertical	2570	25812	30551	8.81
-15	Bkgd Mean	Min	Max	% Modulation
Horizontal	2570	26408	30132	6.90
Vertical	2570	27168	28824	3.10

6.3.3 Depth of Field Test Results

The Depth of Field for each of the three filters in the Intensified Camera Package was determined by Spatial Resolution measurements, in both horizontal and vertical directions, of USAF Target Group 1, Element 6. This group represented the object plane Spatial Resolution limit at just above 10% Contrast.

If the Contrast remained > 10% as the target was translated throughout the range, the resolution was considered to have held at that distance. Depth of Field of the Intensified Camera Package is therefore equal to the range through which the resolution held.

Formaldehyde (Cool Flame): ± 1.0 cm DOF in each direction = 2.0 cm

CH (Hot Flame): ± 1.0 cm DOF in each direction = 2.0 cm

Fused Silica: ± 1.0 cm DOF in each direction = 2.0 cm

Based on the 3.0 cm (± 1.5 cm in each direction) Depth of Field requirement, the measured DOF falls short for each of the filters by ± 0.5 cm in each direction (or a total of 1.0 cm).

Thus, per the stated Depth of Field requirements, the Intensified Camera does not comply. However with approved waiver: CIR-WAI-xxxx, the requirement is met.

APPENDIX A SETUP DATA SHEET FOR RADIOMETER CALIBRATION

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	

Step	Action	Init
1	Obtain all hardware and equipment required for testing. Record model numbers, serial numbers, and calibration dates as necessary in Table 1. *Use caution when handling ESD sensitive materials and optical components.*	
2	Record ambient temperature : _____	
3	Set up the Blackbody Source and the Radiometer Package, using the optical mounting hardware configuration (for either BB or NB Radiometer mode). Roughly align the Blackbody Aperture with the lens at the Radiometer output (in either BB or NB mode) as shown in Figure 4. (More precise alignment will be performed in Steps 9 and 10.)	
4	Set up the support hardware (Blackbody Controller, ±12 VDC Power Supply, Voltmeter. Make the following connections with cables fed through the chamber feed-through port and connected at their respective locations: - Connect the Blackbody to its Controller via vendor supplied cable. - Connect Blackbody Controller to a non-GFI 120 VAC outlet. - Connect Radiometer Package to the ±12 VDC Power Supply via the +12 V, -12 V, and Common Leads on the Radiometer Test Cable. - Connect Radiometer Package to the Voltmeter via the V_{out} and Common Leads on the Radiometer Test Cable. - Connect the DC Power Supply to a 120 VAC outlet. Figure 6 shows a photo of the test configuration in a laboratory setting.	
5	Power up the Blackbody via the Controller. Set the initial blackbody temperature to 700 ºC and aperture = 0.4" for BB / 0.1" for NB. Allow one (1) hour for the Blackbody to reach temperature and stabilize. Record Time On: _____	
6	Set up the ±12 VDC Power Supply to match the Flight Avionics Package ±12 V settings. Connect the Voltmeter to the Power Supply outputs to verify voltages. Record precise +12 V Setting: _____ Record precise -12 V Setting: _____	
7	Power up Radiometer Package via the ±12 VDC Power Supply. Record Time On: _____	
8	Set the horizontal distance from the Radiometer face to the Blackbody Aperture Wheel equal to the appropriate Flight distance (WD = 110 – 9 mm). See Figures 3 and 4.	

Step	Action	Init
9	Log geometric center locations from rail scale (X) and ruler (Y). X: _____ Y: _____	
10	Scan the Radiometer horizontally and the Blackbody Aperture vertically to locate the peak output voltage on the voltmeter. This ensures correct alignment of the devices. Secure Blackbody and Radiometer housing to prevent movement. Log the signal peak location from rail scale (X) and ruler (Y). X: _____ Y: _____	
11	Set-up Complete. If proceeding with other Calibration Steps, leave the hardware powered up. If not, power down all hardware.	

_____ _____

Signature – CFI Test Operator **Date**

_____ _____

Signature – Product Assurance **Date**

Comments:

_____
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APPENDIX B RADIOMETER CALIBRATION PROCEDURE DATASHEET

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	

Step	Action	Init
1	If not already, Power the Blackbody via the Controller. Set the initial blackbody temperature to 1200 ºC. Allow one (1) hour for the Blackbody to reach temperature and stabilize. Record Time On: _____	
2	If not already, Power Radiometer Package via the ±12 VDC Power Supply.	
3	Connect the Voltmeter to the correct V_{out} and power up Voltmeter.	
4	Verify correct Radiometer Set-up ☐ Blackbody normal to detector ☐ Working Distance = 110 - 9 mm ☐ Aligned to peak signal	
5	Prepare a "Radiometer Calibration Data Collection Sheet" (APPENDIX C). **Note:** Separate "Radiometer Calibration Data Collection Sheet" (APPENDIX C) are used for each of the blackbody temperatures and for each of the radiometers.	
6	Cover the Blackbody output.	
7	Record the Radiometer V_{out} from the Voltmeter and record it in the "Closed" row of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
8	Open the Blackbody aperture to the 0.025" setting.	
9	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage" column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
10	Change the Blackbody aperture to the 0.05" setting.	
11	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage" column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
12	Change the Blackbody aperture to the 0.1" setting.	
13	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage" column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
14	Change the Blackbody aperture to the 0.2" setting.	
15	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage" column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
16	Change the Blackbody aperture to the 0.4" setting.	
17	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage"	

Step	Action	Init
	column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
18	Change the Blackbody aperture to the 0.6" setting.	
19	Record the Radiometer V_{out} from the Voltmeter and record in the "Detector Voltage" column of the "Radiometer Calibration Data Collection Sheet" (APPENDIX C).	
20	Decrement the Blackbody Temperature by 100 ºC and allow at least 30 minutes for the Blackbody to reach temperature and stabilize.	
21	Repeat Steps 5 through 20 and record data in the appropriate Detector Voltage Rows on a new "Radiometer Calibration Data Collection Sheet" (APPENDIX C) for each temp.	
22	Continue to decrement the Blackbody Temperature by 100 ºC, allow at least 30 minutes for the Blackbody to reach temperature and stabilize each time and Repeating Steps 6 – 15 until the V_{out} values are equal to the "Closed" values (this typically has occurred around the 300 ºC Blackbody temperature).	
23	Data Collection for Radiometer Calibration has been completed. If continuing with FOV characterization, leave hardware in aligned configuration and proceed to APPENDIX D.	
24	If testing complete, power off all hardware.	

_____ _____

Signature – CFI Test Operator **Date**

_____ _____

Signature – Product Assurance **Date**

Comments:

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APPENDIX C RADIOMETER CALIBRATION DATA COLLECTION SHEET

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	
Ambient Temperature (°C)	
Blackbody Temperature (°C)	

Aperture (Diameter)	Detector Voltage (V)	Notes
0.6”		
0.4”		
0.2”		
0.1”		
0.05”		
Closed		

_____ _____

Signature – CFI Test Operator **Date**

Comments:

_____
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_____
_____
_____
_____

APPENDIX D FIELD OF VIEW CHARACTERIZATION PROCEDURE

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	

Step	Action	Init
1	If not already, Power the Blackbody via the Controller. Set the initial blackbody temperature to 700 ºC and aperture = 0.4" for BB / 0.1 for NB. Allow one (1) hour for the Blackbody to reach temperature and stabilize. Record Time On: _____	
2	If not already, Power Radiometer Package via the ±12 VDC Power Supply.	
3	Connect the Voltmeter to the correct V_{out} and power up Voltmeter.	
4	Verify correct Radiometer Set-up ☐ Blackbody normal to detector ☐ Working Distance = 110 - 9 mm ☐ Aligned to peak signal	
5	Prepare a "FOV Characterization Data Collection Sheet" (APPENDIX E). **Note**: Separate "FOV Characterization Data Collection Sheet" (APPENDIX E) are used for each of the radiometers.	
6	Translate the radiometer on the horizontal rail to the -70 mm position.	
7	Record the "Scale Reading" and Radiometer V_{out} from the Voltmeter on the "FOV Characterization Data Collection Sheet" (APPENDIX E).	
8	Increment the radiometer on the horizontal rail by 10 mm to scan across the FOV.	
9	Record the "Scale Reading" and Radiometer V_{out} from the Voltmeter on the "FOV Characterization Data Collection Sheet" (APPENDIX E).	
10	Repeat Steps 8 and 9 until data are acquired through the +70 mm radiometer position.	
11	Return the radiometer to the center (peak signal) position.	
12	Continue characterization with a scan in the Y direction; lower lab jack(s) to -60 mm.	
13	Record the "Scale Reading" and Radiometer V_{out} from the Voltmeter on the "FOV Characterization Data Collection Sheet" (APPENDIX E).	
14	Increment the blackbody position by 10 mm using the lab jack to scan across the FOV.	
15	Record the "Scale Reading" and Radiometer V_{out} from the Voltmeter on the "FOV Characterization Data Collection Sheet" (APPENDIX E).	
16	Repeat Steps 14 and 15 until data are acquired through the +60 mm blackbody position.	
17	Return the blackbody source to the center (peak signal) position and power down.	

APPENDIX E FOV CHARACTERIZATION DATA COLLECTION SHEET

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	
Ambient Temperature (°C)	
Blackbody Settings	**T: _____ °C Aperture: _____ ”**

X Radiometer Position (mm)	Scale Reading (mm)	Detector Voltage (V)	Notes
-70			
-60			
-50			
-40			
-30			
-20			
-10			
0			
10			
20			
30			
40			
50			
60			
70			

_____ _____

Signature – CFI Test Operator **Date**

_____ _____

Signature – Product Assurance **Date**

FOV CHARACTERIZATION DATA COLLECTION SHEET (Part 2)

Date & Time (Start / Complete)	
Test Engineer	
Detector (BB or NB)	
Ambient Temperature (°C)	
Blackbody Settings	**T: _____ °C Aperture: _____ ”**

Y Radiometer Position (mm)	Scale Reading (mm)	Detector Voltage (V)	Notes
-70			
-60			
-50			
-40			
-30			
-20			
-10			
0			
10			
20			
30			
40			
50			
60			
70			

_____ _____

Signature – CFI Test Operator **Date**

_____ _____

Signature – Product Assurance **Date**

APPENDIX F SNR DATA COLLECTION WORKSHEET

Detector	Temp (°C)	Aperture (in. dia.)	G(T) (W/cm^2)	DC Volts	AC Volts	Filename	Date/Time

APPENDIX G ACRONYMS AND ABBREVIATIONS

BB	Broadband
CIR	Combustion Integrated Rack
COTS	Commercial Off The Shelf
DC	Direct Current
ECO	Engineering Change Order
ERD	Engineering Requirements Document
ESD	Electrostatic Discharge
FCF	Fluids and Combustion Facility
FLEX	Flame Extinguishment Experiment
CFI	Cool Flames Investigation
FOV	Field of View
IR	infrared
ISS	International Space Station
MDCA	Multi-user Droplet Combustion Apparatus
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NB	Narrowband
SNR	Signal to Noise Ratios
SpaceDOC	Space Flight Systems Development and Operations Contract
VAC	Volts (alternating current)
VDC	Volts (direct current)